On Adversarial Attacks and Defenses in Deep Learning: A Systematic Study on MNIST

Wang Wenxuan | Student ID: 1397228
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Code and data: `https://github.com/wwx0015/cds521-adversarial-mnist`

Abstract

Deep neural networks, despite high accuracy on many vision benchmarks, can be fooled by imperceptibly small perturbations crafted by adversaries. This dissertation surveys the threat model, taxonomy, and defenses of *adversarial machine learning*, and reports a controlled empirical study on MNIST. Using a four-layer convolutional classifier (421 K parameters) trained on an NVIDIA RTX 2050 GPU, we (i) implement the Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD) as white-box L_∞ attacks, (ii) compare them against a Madry-style PGD adversarially trained variant, and (iii) analyze class-level vulnerability and compute cost. At $\varepsilon = 0.3$, PGD-20 reduces the baseline's accuracy from **98.86**% to effectively **0**%, whereas the adversarially trained model retains **79.00**%. Adversarial training costs ≈ 5.2 pp in clean accuracy and a 6.6× training wall-clock overhead, and robustness is demonstrably non-uniform across digit classes. These results are consistent with the central finding of Madry *et al.* (2018) at small scale, and motivate our discussion of certified defenses and physically realizable attacks.

1 Introduction

Szegedy et al. [11] discovered that state-of-the-art image classifiers misclassify inputs perturbed by modifications imperceptible to humans. Goodfellow et al. [5] argued the phenomenon stems from the (near-)linear geometry of deep networks and proposed the **Fast Gradient Sign Method (FGSM)**, a single-step attack. Madry et al. [7] reframed robust learning as a min–max optimization

$$\min_{\theta} \; \mathbb{E}_{(x,y)\sim\mathcal{D}}\left[\max_{\|\delta\|_\infty \le \varepsilon} \mathcal{L}(f_\theta(x+\delta),\, y)\right], \qquad (1)$$

and introduced **Projected Gradient Descent (PGD)**—a multi-step variant of FGSM—as a strong first-order adversary. The field has since matured into a formal taxonomy codified by Vassilev et al. [8]. This paper reviews that taxonomy, reproduces FGSM and PGD on MNIST, evaluates PGD-based adversarial training, and supplements the standard accuracy-vs-budget plot with class-level and computational-cost analyses (Figures 1–3, Tables 1–2).

2 Theoretical Background

Adversarial examples. Given a classifier f_θ and a sample (x, y), an *adversarial example* is $x' = x + \delta$ with $\|\delta\|_p \le \varepsilon$ such that $f_\theta(x') \ne y$. The *budget* ε bounds perceptibility; $p \in \{0, 2, \infty\}$ selects the geometry (pixel-count, Euclidean, uniform per-pixel). We adopt $p = \infty$ throughout, following Goodfellow et al. [5].

White-box vs. black-box. A *white-box* adversary knows θ and can compute $\nabla_x \mathcal{L}$; FGSM and PGD belong here. A *black-box* adversary only queries the model as an oracle, yet *transferability* [9] makes white-box examples surprisingly effective across unseen models.

FGSM. Linearizing $\mathcal{L}$ around x, the worst-case L_∞ perturbation of budget ε is $\delta^\star = \varepsilon\,\mathrm{sign}\big(\nabla_x \mathcal{L}\big)$, giving $x' = \mathrm{clip}_{[0,1]}(x + \delta^\star)$. This is exact for a truly linear $\mathcal{L}$ and a useful first-order approximation otherwise.

PGD. PGD iterates the FGSM step T times with stride α, projecting back to the ε-ball (and pixel range) after each step. It is widely regarded as the strongest *first-order* adversary [7].

Defense taxonomy. Defenses fall into three families: (i) *adversarial training*—augmenting the loss with perturbed inputs, approximating the inner max of Eq. (1); (ii) *input preprocessing* such as feature squeezing [12], which attempts to remove adversarial signal; and (iii) *certified defenses* such as randomized smoothing [3], which provide provable robustness radii. Defenses that merely obscure $\nabla_x \mathcal{L}$—"gradient masking"—were shown by Athalye et al. [1] to give a false sense of security.

3 Application-Based Analysis

Real-world scenarios. (i) *Autonomous driving:* Eykholt et al. [4] fooled a road-sign classifier with black-and-white stickers on stop signs—a physically realizable attack robust to viewpoint and distance. (ii) *Face recognition:* Sharif et al. [10] 3-D-printed eyeglass frames that caused commercial face verifiers to impersonate chosen identities. (iii) *Malware detection:* Grosse et al. [6] evaded neural malware classifiers by appending benign bytes, demonstrating that the threat is not confined to vision.

Threat model. Following Vassilev et al. [8], an adversary is characterized by four axes: *goal* (targeted/untargeted), *knowledge* (white-/grey-/black-box), *capability* (L_p budget, query budget, physical constraints), and *access* (train-time poisoning vs. test-time evasion). This study focuses on **white-box, test-time, untargeted, L_∞-bounded evasion**.

Pipeline. (1) train or obtain a target classifier; (2) for

each test input, compute $\nabla_x \mathcal{L}$ via back-propagation; (3) synthesize δ under the chosen budget; (4) measure attack success; (5) for defense, retrain with the attack in the loop, approximating Eq. (1).

4 Implementation

We train a CNN (two 3×3 conv + max-pool blocks, 128-unit FC head; 421 K parameters) on MNIST for 3 epochs with Adam ($\eta = 10^{-3}$, batch 128). The full 10 000-sample test set is used for evaluation. Seven ε values are swept, $\varepsilon \in \{0, .05, .10, .15, .20, .25, .30\}$. Four model × attack conditions are compared: (standard, adv-trained) × (FGSM, PGD-20). Evaluation PGD follows the Madry protocol: $T = 20$ steps, random start, step size $\alpha = 2.5\varepsilon/T$ (so the attack can reach the full ε-ball). The adversarially trained model is obtained via PGD-7 with random initialization, $\alpha = \varepsilon/4$, $\varepsilon = 0.3$, for 5 epochs (Madry-style). All experiments run on an NVIDIA GeForce RTX 2050 (4 GB). Key FGSM pseudocode:

```
def fgsm(model, x, y, eps):
    x = x.clone().detach().requires_grad_(True)
    loss = F.cross_entropy(model(x), y)
    grad = torch.autograd.grad(loss, x)[0]
    return (x + eps*grad.sign()).clamp(0, 1).
        detach()
```

5 Experimental Results

Clean accuracy. The baseline reaches **98.86**% on the 10k test set; the PGD adversarially trained variant attains **93.71**%—a 5.15 pp tax for robustness.

Qualitative visualization. Figure 1 shows two MNIST digits, their FGSM perturbations at $\varepsilon = 0.25$, and the resulting adversarial images. The perturbations resemble low-amplitude noise, yet both predictions flip with high confidence (7→3 and 2→6).

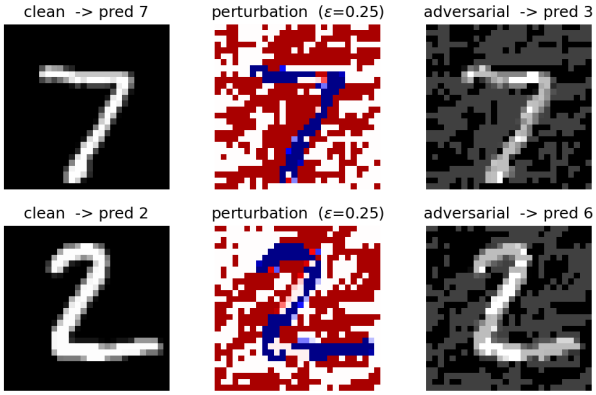

Figure 1: Original, perturbation, and FGSM adversarial examples ($\varepsilon = 0.25$). The perturbation (center) amplified by a divergent color map; the resulting adversarial image (right) is visually close to the original yet flips the model's prediction.

Robustness sweep. Table 1 reports accuracy across the budget grid; Figure 2(a) plots the same data, and Figure 2(b) breaks the standard-model accuracy at $\varepsilon = 0.2$ down by digit class.

ε	FGSM / std	PGD / std	FGSM / rob	PGD / rob
0.00	98.86	98.86	93.71	93.71
0.05	95.36	93.84	92.09	91.92
0.10	84.47	69.48	90.47	89.92
0.15	60.38	18.84	89.00	87.97
0.20	31.71	0.47	87.80	85.54
0.25	11.19	0.00	86.92	82.89
0.30	2.56	0.00	86.13	79.00

Table 1: Test accuracy (%) of the standard and PGD adversarially trained CNN under FGSM and PGD-20 across the L_∞ budget grid. PGD saturates at 0% on the standard model by $\varepsilon = 0.25$.

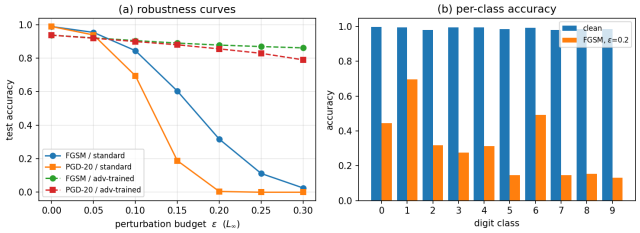

Figure 2: (a) Robustness vs. perturbation budget ε for the two models and two attacks. (b) Per-class accuracy at $\varepsilon = 0.2$ for the standard model, revealing pronounced heterogeneity across digits.

Error structure. Figure 3 contrasts the confusion matrix before and after FGSM attack at $\varepsilon = 0.2$ on the standard model. The clean diagonal collapses, and errors concentrate into a few "attractor" columns (notably 1, 6, 9), often classes differing from the true class by a single stroke.

Compute cost. Table 2 contrasts the wall-clock cost of the three attacks on 1 024 images on the RTX 2050. PGD-20—the standard evaluation adversary—is 17.5× more expensive than FGSM, making attack strength a routine engineering trade-off during model assurance.

Attack	Iterations	Time (s) / 1024	Relative
FGSM	1	0.047	1.00×
PGD	7	0.304	6.43×
PGD	20	0.829	17.50×

Table 2: Wall-clock cost of each attack on 1 024 MNIST test images (NVIDIA RTX 2050).

Findings. (i) PGD dominates FGSM on the standard model at every ε; the gap grows from 1.52 to 41.54 pp between $\varepsilon = 0.05$ and 0.15 and closes only because PGD saturates at 0% by $\varepsilon = 0.25$ (Table 1). (ii) Adversarial training retains 79.00% accuracy under PGD-20 at $\varepsilon = 0.3$ while the standard model collapses to 0%—a result consistent with the central finding of Madry et al. [7] at small scale. (iii) Robustness is **non-uniform across classes** (Figure 2b): at $\varepsilon = 0.2$, digit 1 retains 69.5% accuracy while digit 9 falls to 13.0%. Digits with more visual "competitors" are attacked first, supporting the

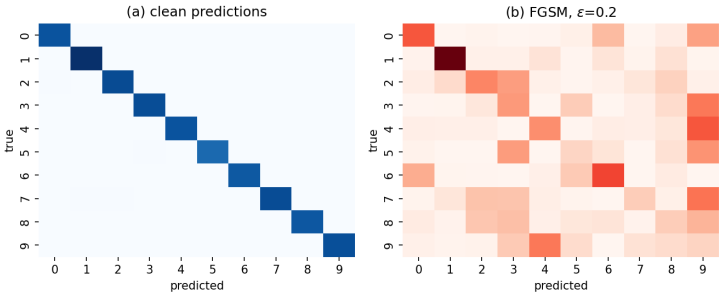

Figure 3: Confusion matrices before (blue) and after (red) a white-box FGSM attack at $\varepsilon = 0.2$ on the standard model. Under attack, errors concentrate into a few "attractor" classes rather than scattering uniformly, indicating structured rather than random failure.

view that adversarial error concentrates along decision-boundary geometry rather than distributing uniformly. (iv) PGD-20's 17.5× compute cost over FGSM (Table 2) matters when adversarial evaluation is a routine step in a CI pipeline. (v) Adversarial training took 112.8 s vs. 17.0 s for clean training, a 6.6× total overhead (≈4× per epoch from the PGD-7 inner loop plus 1.67× from the longer schedule).

6 Ethical Considerations

Adversarial ML is **dual-use**. Offensively, the same techniques that fool a stop-sign classifier can (i) enable impersonation against biometric checkpoints, (ii) craft deepfakes that evade forensic classifiers, or (iii) bypass spam and malware filters. Publishing attack code therefore risks arming malicious actors. Yet *withholding* attacks leaves defenders blind: Vassilev et al. [8] explicitly recommend disclosure because "security through obscurity" systematically favors attackers. A responsible stance—adopted here—reproduces published attacks on toy datasets while foregrounding defense. A second ethical axis concerns **fairness**: reporting only *average* robustness can hide the fact that some classes or subgroups are markedly more vulnerable than others, so fairness harms may compound under adversarial attack [8]. Our per-class analysis (Figure 2b) is the MNIST analogue of this concern—robustness gaps spanning 56.5 pp between the easiest and hardest digit at $\varepsilon = 0.2$—motivating the reporting of *worst-class* robustness alongside average accuracy.

7 Future Directions

Three frontiers appear most promising. **(1) Certified defenses.** Cohen et al. [3] convert a base classifier into a provably robust one by majority-voting over Gaussian-noised inputs; the certificate scales favorably to ImageNet. **(2) Physically realizable attacks.** Beyond digital perturbations, the *Expectation-over-Transformation* framework produces adversaries robust to real-world imaging pipelines [2], motivating sensor-level and deployment-time defenses. **(3) Foundation-model robustness.** Large pretrained models exhibit emergent behaviors under adversarial prompting [13]; extending the L_∞ threat model to text, speech, and multi-modal inputs is an open and socially consequential direction.

8 Conclusion

We surveyed the landscape of adversarial machine learning and empirically reproduced FGSM, PGD, and Madry-style adversarial training on MNIST under a modest compute budget. The quantitative picture (Tables 1–2, Figures 1–3) confirms that (i) adversarial vulnerability is severe for standardly trained networks, (ii) adversarial training is an effective, if computationally costly, remediation, (iii) the benefit is non-uniform across classes, and (iv) strong attacks are 24× more expensive than weak ones. Trustworthy AI will require combining such empirical defenses with certified guarantees, physical-world evaluation, and careful dual-use ethics.

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